The evolution of environmental impacts of platform chemicals linked to climate policies

A. Nabera,[a] A. J. Martín,[a] R. Istrate,[b] J. Pérez-Ramírez,[a] and G. Guillén-Gosálbez[a]

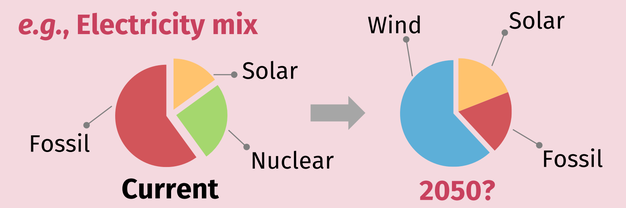
[a]Institute for Chemical and Bioengineering, ETH Zürich, Switzerland.
[b]Institute of Environmental Sciences (CML), Leiden University, The Netherlands.
Email: abhinandan.nabera@chem.ethz.ch

Introduction

- The chemical industry releases 5.6 Gt CO_2-eq yr^{-1} (10% of global emissions) due to its reliance on fossil fuels.[1]
- Switching to low-carbon hydrogen derived from electrocatalytic solar or wind-powered water electrolysis can substantially reduce emissions for ammonia production.
- Current environmental studies do not consider the future evolution of power, materials, and transportation sectors under different climate policy scenarios.

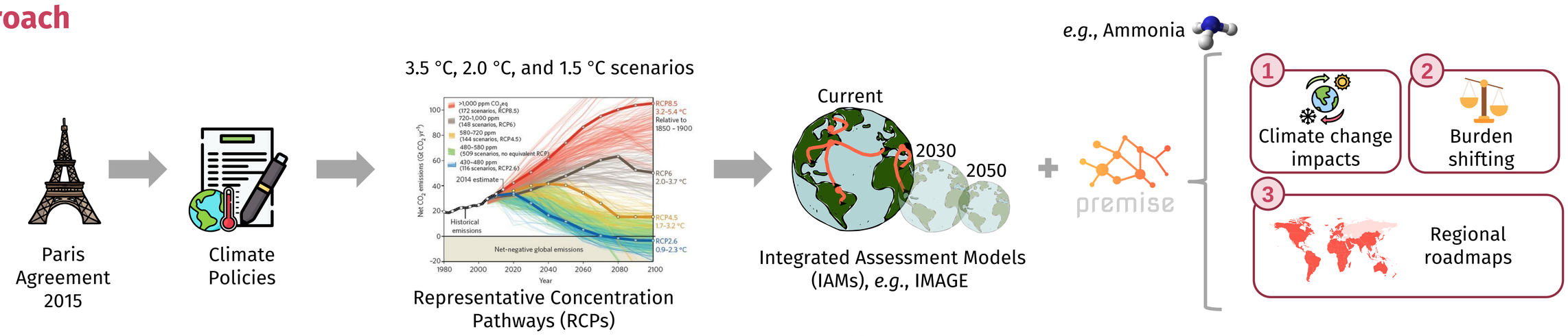
Goal

- Demonstrate how climate policies can synergistically drive sustainability in the industry.
- **How do the different production routes compete in the future?**

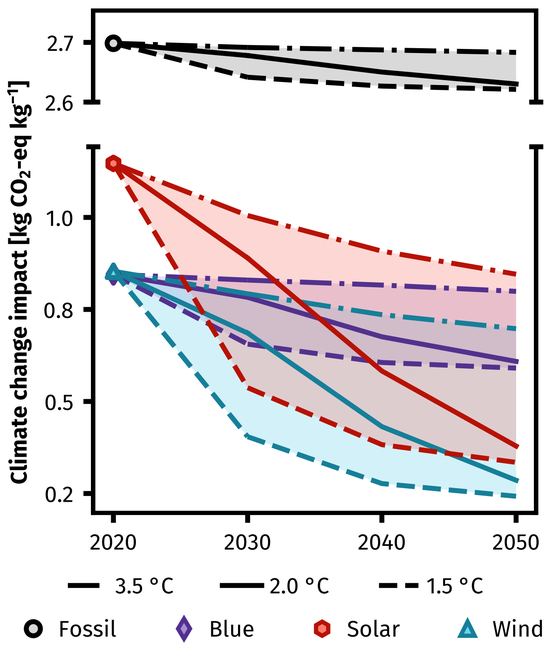


Prospective life cycle assessment!

Approach



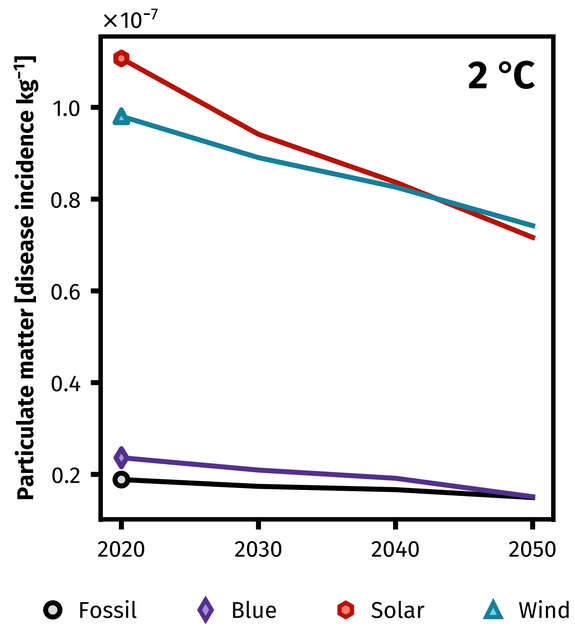
1 Climate change impacts



- The green routes (solar and wind) could produce ammonia with 90% fewer emissions in the 1.5 °C scenario compared to the emissions from the fossil scenario in 2020.
- The reduction is primarily driven by decreased consumption and further decarbonisation of renewable electricity.
- Fossil-based thermocatalytic scenarios show a reduction of less than 10%, even in the best case, *i.e.*, the 1.5 °C scenario.
- Blue ammonia, while currently having a smaller impact than green routes, is projected to be outperformed by green ammonia in future scenarios.

90% reduction in emissions can be achieved by 2050

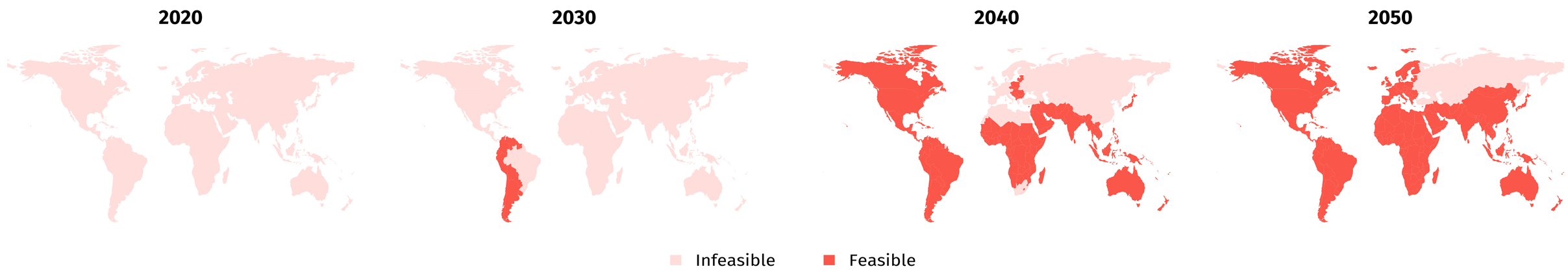
2 Burden shifting



- Analysis extended beyond carbon footprint to assess the potential risk of burden shifting, *i.e.*, worsening other impacts while attempting to combat climate change.
- Green routes for ammonia exhibit higher particulate matter impacts compared to fossil routes, indicating burden shifting.
- Renewable electricity, particularly from solar panels and wind turbines, is identified as a primary source of particulate matter impacts.
- The severity of the burden shifting tends to become less pronounced over time for the green routes.

Burden shifting exists, albeit less pronounced

3 Regional roadmaps (solar ammonia, 2 °C scenario)



- We define feasibility as achieving an 80% reduction in climate change impact compared to the fossil-based scenario in 2020.
- The first region expected to meet this stringent criterion by 2030 is South America, excluding Brazil. By 2050, all regions are anticipated to achieve the established goal, except for Russia, Turkey, and Central Asia.
- The primary reason for the variability among regions is the diverse power mixes deployed worldwide.

Regional feasibility varies by power mix

Conclusions and Outlook

- Climate policies will significantly enhance the environmental benefits of green electrocatalytic routes.
- These distilled insights provide tools for researchers, policymakers, industry leaders, and investors to encourage the green production of chemicals.

References [1] *Curr. Opin. Chem. Eng.* **2023**, 39, 100881

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